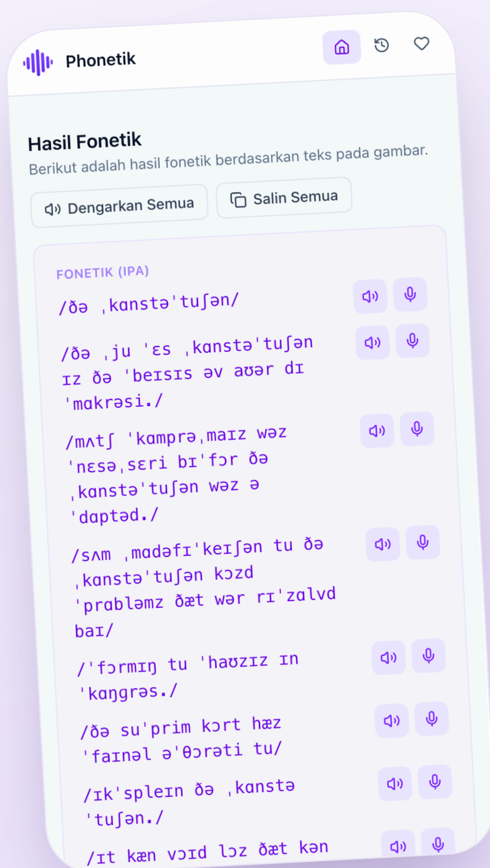
Phonetik

CASE STUDY

# From a Photo to Perfect Pronunciation

An AI pipeline that reads printed English text, transcribes it into IPA phonetics, and coaches the reader to speak it correctly.

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phonetik-translation-platform.vercel.app

OVERVIEW

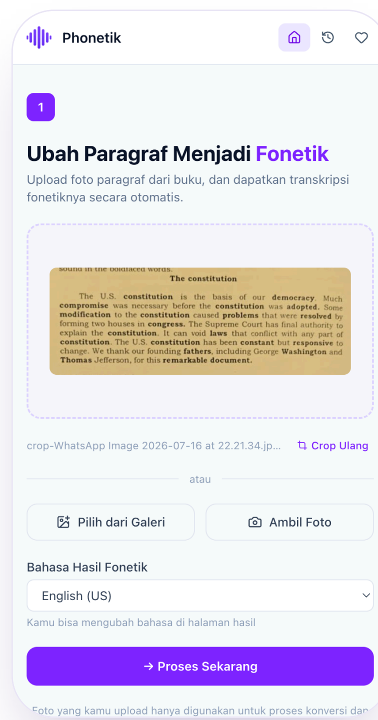
# What Phonetik Does

Phonetik turns a photo of printed English text into an IPA phonetic transcription, then coaches the reader to pronounce it correctly — record, score, and get line-by-line feedback, in one flow.

**01** 
## Capture
Photograph a paragraph from a book, worksheet or article.

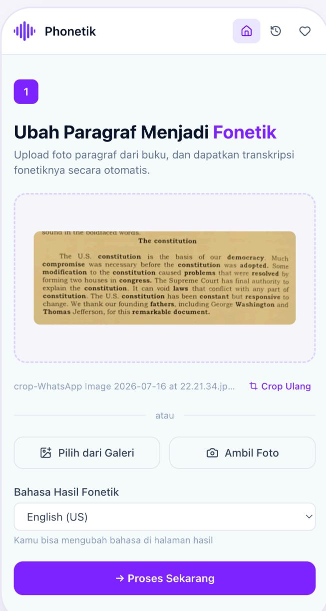
**02** 
## Transcribe
A vision model reads the text and converts every line into IPA.

**03** 
## Practice
Read it aloud and get an instant pronunciation score.



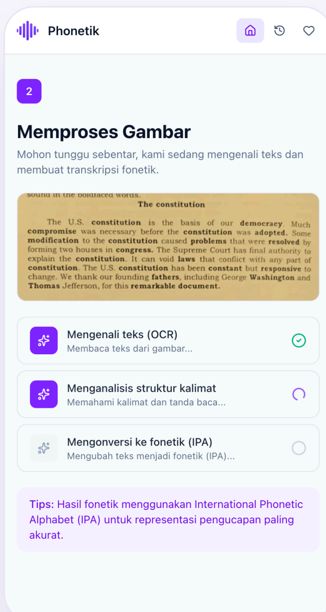
PRODUCT WALKTHROUGH

# The Learning Loop, Screen by Screen



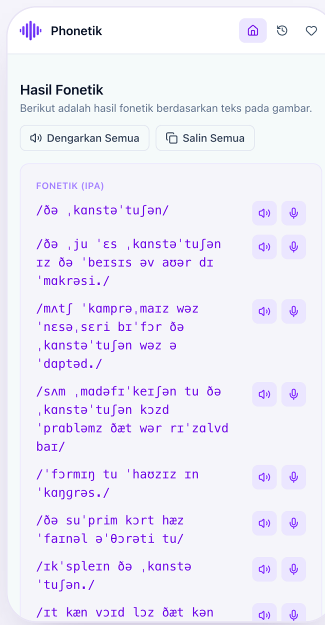

## Upload

Snap or upload a photo of the paragraph.



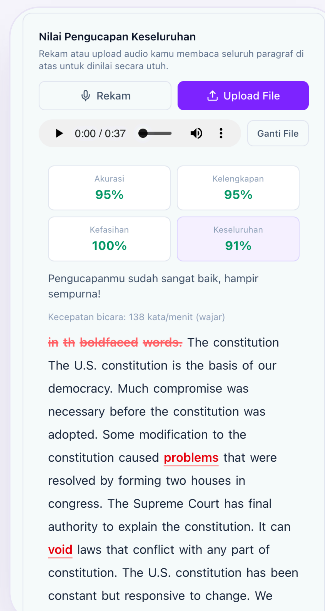

## Process

OCR, sentence analysis and IPA conversion run as visible steps.




## Transcribe

Every line gets a broad IPA transcription to play or copy.




## Practice

Record a reading and get scored on the spot.

AI PIPELINE · 01

# Generating Phonetics With a Vision Model

A Supabase Edge Function sends the photo to Gemini 2.5 Flash via OpenRouter, with a prompt pinned to one phonemic convention — the Longman American English key — down to stress marks and syllables.

### Structured output

Strict JSON only — original text plus per-line IPA, no markdown fences.

### Self-healing retries

Up to 3 attempts on invalid, empty or malformed JSON before failing.

### Symbol normalization

British-leaning symbols the model drifts into are rewritten to General American equivalents post-generation.

### Cost guardrails

Usage capped at 30 conversions per user per rolling 24 hours.

RESPONSE SHAPE

```
{
  "originalText": "...",
  "lines": [
    { "text": "...",
      "ipa": "/ðə kənstəˈtuʃən/" }
  ]
}
```

AI PIPELINE · 02

# Scoring Real Speech Against the Transcript

The recording is transcribed by Groq's Whisper large-v3-turbo, then diffed word-by-word against the expected line using the same edit-distance alignment behind Word Error Rate.

### Accuracy

Correct words ÷ words attempted.

### Completeness

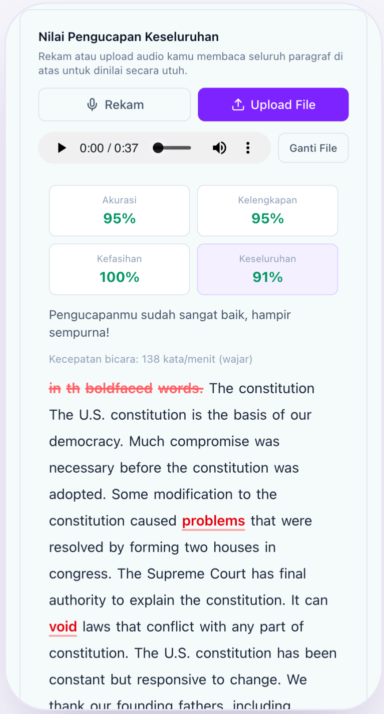
Words attempted ÷ total words.

### Fluency

Speaking pace vs. an ideal 130 wpm.

### Overall

Correct words ÷ total words.



ARCHITECTURE

# System Architecture

**apps/platform**
TanStack Start · React 19

**apps/vocab**
TanStack Start · React 19

**apps/admin**
TanStack Start · React 19

↓

**packages/ui — shared component library (Tailwind CSS v4)**

↓

**Supabase**
Postgres + RLS · Auth · Storage · Edge Functions (Deno)

**apps/api**
Hono + Prisma — typed REST layer

pnpm workspaces · TypeScript · Biome · Docker · GitHub Actions · Vercel

ENGINEERING

# Engineering for Real Users

### Row-level security

Every table and the private audio bucket are scoped by Postgres RLS keyed to auth.uid() — a user only ever reads their own data.

### Rate-limited AI spend

Rolling 24-hour caps — 30 OCR conversions, 50 pronunciation attempts per user — guard against runaway model costs.

### Resilient model calls

Both pipelines validate response shape and retry up to 3 times before surfacing an error to the user.

### Consistent phonetic output

A normalization layer keeps IPA symbols consistent across model versions and drift.

Phonetik

# Let's Talk

Phonetik is a solo-built, production-shaped project — typed end-to-end, deployed on Vercel and Supabase.

- 4 apps, 1 monorepo
- 2 AI providers
- RLS on every table

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